

Lab 02

Operationalization, Variables & Measurement

Walk-through (15 points)

Instructions

Identify Types of Variables

For each of the variables listed below, highlight (use yellow) whether it is **quantitative** (values reflect a count or measurement) or **qualitative** (values correspond to category membership). For each quantitative, you also need to highlight (use yellow) whether it is **continuous** (can take on any value; e.g., decimal values) or **discrete** (integer or whole number values only).

1. List of descriptive characteristic variables

Table 1

Superman Characteristics Variables

Variable	Description	Example Values	Qual or Quant	Discrete or Continuous
type	Type of media: 1=Movie, 2=TV Series, 3=Serial/Short Film	1, 2		
first_appear	Year first debuted (YYYY)	1978, 2025		
clark_height	Height of Superman actor in meters (to 2 decimal places)	1.93, 1.85		
height_grp	Whether or not Superman actor is over 6ft tall: 1=Yes ($\geq 1.83\text{m}/6'0''$), 2=No ($< 1.83\text{m}/6'0''$)	1, 2		
lois_height	Height of Lois Lane actor in inches (rounded up)	63, 68		
height_diff	Height difference (Clark's height - Lois's height) in feet (to 2 decimal places)	0.98, 0.72		

2. List of enjoyment/quality variables

Table 2*Superman Quality Assessment Variables*

Variable	Description	Example Values	Qual or Quant	Discrete or Continuous
rt_critics_score	Rotten Tomatoes critics score (0-100%)	94, 56		
rt_status	Rotten Tomatoes status (1=Fresh, 2=Certified Fresh, 3=Rotten)	1, 3		
rt_audience_score	Rotten Tomatoes audience score (0-100%)	88, 62		
lb_avg_stars	Letterboxd average rating (1-5 stars)	4.2, 3.8		
lb_reviews	Letterboxd number of reviews	1250, 890		
metacritic_score	Metacritic score (0-100)	78, 52		
emotion	Primary emotion response while watching (1=fear, 2=joy, 3=sadness, 4=anger, 5=disgust, 6=anxiety)	2, 2		
emo	Emotional Impact Subscale of Subjective Movie Enjoyment Scale (SMES) sum of 4-items (1-5 scale) measuring emotional response	16, 18		
aes	Aesthetic Appeal Subscale of Subjective Movie Enjoyment Scale (SMES) sum of 3-items (1-5 scale) measuring visual appeal	12, 14		
cog	Cognitive Engagement Subscale of Subjective Movie Enjoyment Scale (SMES) average of 4-items (0-7 scale) measuring cognitive stimulation	6.3, 4.7		

None

Discussion Question: In the above dataset we have several variables that represent different operationalizations of the same or similar behavior/characteristics.

*How might these **different operationalizations** (i.e., different measures of height or quality) influence generalizability or how we interpret the results?*

In-Class (30 points)

Instructions

Find a partner and work on the next few sections together!!

Part A. Creating conceptual definitions

In **Part A** below, you will practice identifying **conceptual definitions** for different psychological constructs.

Conceptual definition: describes the behaviors or internal processes that might make up a psychological construct, along with how it relates to other variables.

1. EXAMPLE: Binge-watching

Conceptual definition: **Watching multiple episodes of a TV show in a single sitting, often leading to the neglect of other activities or responsibilities.**

2. Physical attractiveness

Conceptual definition:

3. Social Media Addiction

Conceptual definition:

4. Risky behavior

Conceptual definition:

5. Emotional intelligence

Conceptual definition:

Instructions**Part B. Operationalizing variables**

In the next section, you and your partner will practice **creating operational definitions** by identifying both **self-report** and **observational/behavioral** measures for each given variable.

Operational definition: a definition of a variable in terms of precisely how it is to be measured.

1. EXAMPLE: Binge-watching

Self-report: **Participants rate how many times in the last 6-months they have binged watched television (defined as watching 3+ consecutive episodes or 2+ hours of continuous viewing of the same show with breaks <15 minutes between episodes) on a scale from 1 (never) to 10 (all the time).**

Observational/Behavioral: **A researcher monitors participants actual TV/streaming viewing behavior over a 6-month period by installing a TV monitoring box/device that automatically records their TV viewing patterns. Researchers count the number of times the device shows 3+ hours of continuous viewing the same program.**

2. Fear experienced while watching a scary movie

Self-report:

Observational/Behavioral:

3. Ability to perceive the difference between colors

Self-report:

Observational/Behavioral:

4. Current Relationship Status

Self-report:

Observational/Behavioral:

5. Willingness to engage in risky behavior

Self-report:

Observational/Behavioral:

Instructions**Part C. Identifying variables from research questions**

Finally, in the next section you and your partner will work to **identify and define variables within existing research questions**.

For each research question, first identify the *target population* is that is being studied. Second, identify the *grouping/independent variable (IV)* and the *dependent variable (DV)* described in the research question. Next, provide an *operational definition* for each variable. Finally, classify each variable as either *quantitative or qualitative*. For quantitative variables, also specify if they are *continuous or discrete*.

1. **EXAMPLE: Is there a relationship between the amount of television watched (hours per day) and happiness (self-report survey 1-10 scale) among adult women?**

Target population: **Adult Women**

- Grouping/Independent Variable:

Grouping/Independent Variable: **Amount of television watched**

IV Operationalization: **Hours per day**

IV type (quantitative or qualitative): **quant**

If IV quant (continuous or discrete): **discrete (i.e., hours)**

- Dependent Variable:

Dependent Variable: **Happiness**

DV Operationalization: **self-report survey 1-10 scale**

DV type (quantitative or qualitative): **quant**

If DV quant (continuous or discrete): **discrete**

2. Is there a relationship between weather conditions and depression in Nebraskan high school students?

Target population:

• Grouping/Independent Variable:

Grouping/Independent Variable:

IV Operationalization:

IV type (quantitative or qualitative):

If IV quant (continuous or discrete):

• Dependent Variable:

Dependent Variable:

DV Operationalization:

DV type (quantitative or qualitative):

If DV quant (continuous or discrete):

3. Are wealthy, upper-class people more likely to engage in unethical behavior?

Target population:

- Grouping/Independent Variable:

Grouping/Independent Variable:

IV Operationalization:

IV type (quantitative or qualitative):

If IV quant (continuous or discrete):

- Dependent Variable:

Dependent Variable:

DV Operationalization:

DV type (quantitative or qualitative):

If DV quant (continuous or discrete):

4. Does daily social media use affect sleep quality in teenagers?

Target population:

- Grouping/Independent Variable:

Grouping/Independent Variable:

IV Operationalization:

IV type (quantitative or qualitative):

If IV quant (continuous or discrete):

- **Dependent Variable:**

Dependent Variable:

DV Operationalization:

DV type (quantitative or qualitative):

If DV quant (continuous or discrete):

5. Does exposure to non-violent, cooperative video games increase prosocial behaviors in children?

Target population:

- **Grouping/Independent Variable:**

Grouping/Independent Variable:

IV Operational definition:

IV type (quantitative or qualitative):

If IV quant (continuous or discrete):

- Dependent Variable:

Dependent Variable:

DV Operationalization:

DV type (quantitative or qualitative):

If DV quant (continuous or discrete):

Take-Home (15 points)

Instructions

Developing empirically testable research hypotheses

Develop a research idea into a working hypothesis based on some current news headlines. There are three news articles that you can use for this assignment. You will need to answer the questions and develop a research hypothesis for **two** of them.

Please do not forget to indicate which article you selected below.

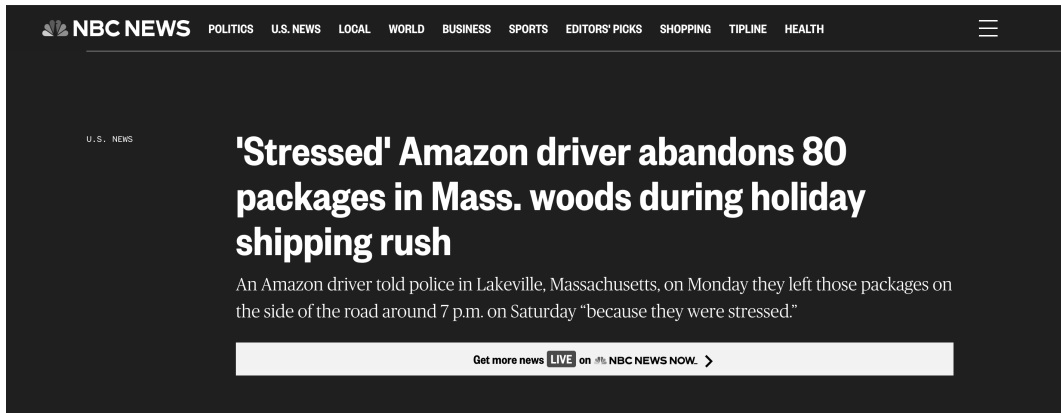


Figure 1: Article 1

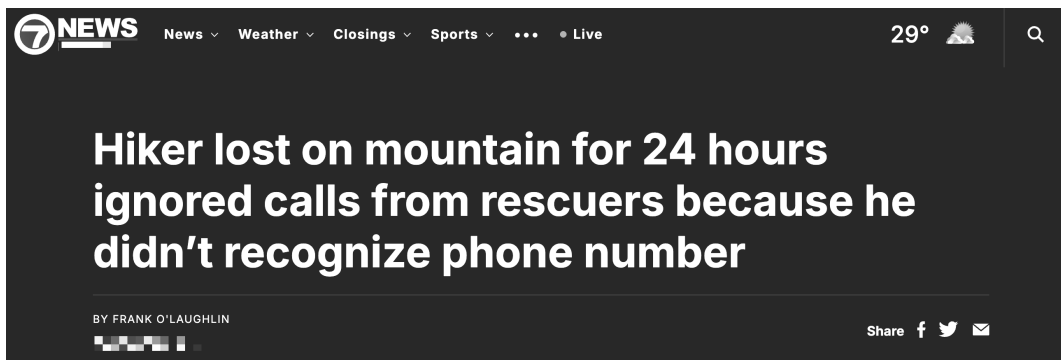


Figure 2: Article 2

MISSOURI **Missouri man must watch 'Bambi' monthly as part of poaching punishment, judge rules**

By Kaitlyn Schallhorn · Fox News

Figure 3: Article 3

1. Select your first news headline and write it below (ie., Article 1 or Article 2 or Article 3)

a. State your initial, general research question based on the article:

Research question:

- b. List and conceptually define all the behaviors/characteristics/concepts/ideas that need to be defined and/or measured to answer your research question.

Behaviors/characteristics/concepts/ideas:

- c. Define an grouping/independent and a dependent variable that can be used to test your research question:

- Grouping/Independent Variable:

Grouping/Independent Variable:

IV Operationalization:

IV type (quantitative or qualitative):

If IV quant (continuous or discrete):

- Dependent Variable:

Dependent Variable:

DV Operationalization:

DV type (quantitative or qualitative):

If DV quant (continuous or discrete):

- d. Identify the target population

Target population:

- e. Finally, state a **research hypothesis** that details your expectation/prediction about the relationship between the grouping/independent and dependent variable.

Research hypothesis:

2. Select your second news headline and write it below (*ie., Article 1 or Article 2 or Article 3*)

- a. State your initial, general research question based on the article:

Research question:

- b. List and conceptually define all the behaviors/characteristics/concepts/ideas that need to be defined and/or measured to answer your research question.

Behaviors/characteristics/concepts/ideas:

- c. Define an grouping/independent and a dependent variable that can be used to test your research question:

- Grouping/Independent Variable:

Grouping/Independent Variable:

IV Operationalization:

IV type (quantitative or qualitative):

If IV quant (continuous or discrete):

- Dependent Variable:

Dependent Variable:

DV Operationalization:

DV type (quantitative or qualitative):

If DV quant (continuous or discrete):

d. Identify the target population

Target population:

e. Finally, state a **research hypothesis** that details your expectation/prediction about the relationship between the grouping/independent and dependent variable.

Research hypothesis: